SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone Presentation | IBM Developer Skills Network

Candidate: Muhammad Qasim **Course:** IBM Data Science Capstone **Track:** Commercial Space Launch Optimization

Project GitHub Repository: https://github.com/QASIM07J/IBM-Applied-Data-Science-Capstone

PRESENTATION OUTLINE & STRUCTURE (PPT SLIDES 2 - 5)

Overview & Context

Executive Summary (PPT Slide 3)

SpaceX advertises Falcon 9 launches at $62 million, substantially below the typical commercial market rate of $165+ million. This cost advantage stems from recovering and reusing the rocket first stage. This project models and predicts whether the first stage lands successfully based on launch parameters, establishing commercial launch cost estimation intelligence.

Project Background & Context (PPT Slide 4)

Commercial competitors must determine the likelihood of a successful booster landing to price competitive bids accurately. By synthesizing REST API data, Wikipedia web records, relational databases, geospatial maps, and predictive algorithms, this study quantifies core reusability determinants across all flight profiles.

Deliverable Outline & Evaluation Rubric Mapping

- **Section 1 (Methodology):** SpaceX API requests, HTML scraping, Data Wrangling, EDA (Visualization & SQL), Folium, Dash, and Classification Pipelines (Rubrics 1.5 - 1.10).
- **Section 2 (EDA Results):** Scatter plots, orbital bar charts, yearly trendlines, and 10 SQL database evaluations (Rubrics 1.11 - 1.12).
- **Sections 3, 4 & 5 (Interactive & ML):** Folium map pins and proximities, Plotly Dash dashboard interactions, model benchmarks, confusion matrix, conclusions, and appendix (Rubrics 1.13 - 1.15).

SECTION 1: METHODOLOGY - DATA COLLECTION (PPT SLIDES 6 - 9)

ETL Architecture

API & Web Scraping GitHub Code: https://github.com/QASIM07J/IBM-Applied-Data-Science-Capstone

Methodological Framework Overview (PPT Slide 6 & 7)

Data was gathered across two primary pipelines: querying the official SpaceX API for raw telemetry and scraping Wikipedia for supplemental records. Collected observations underwent structured parsing, type conversions, missing-value imputation, and feature engineering prior to relational storage and modeling.

SpaceX REST API Pipeline & Flowchart (PPT Slide 8 | Rubric 1.5)



Iterated through endpoints `/launches/past`, `/payloads`, `/launchpads`, and `/cores`. Extracted 90 verified Falcon 9 launch records while pruning developmental Falcon 1 flights.

Wikipedia Web Scraping Pipeline & Flowchart (PPT Slide 9 | Rubric 1.6)



Parsed tabular HTML launch tables, extracted flight numbers, launch sites, customer payloads, and landing targets to cross-verify API telemetry and resolve unrecorded historical values.

SECTION 1: METHODOLOGY - PROCESSING, EDA, ANALYTICS & ML (PPT SLIDES 10 - 15)

Analytics Framework

EDA, Folium & ML GitHub Repositories: https://github.com/QASIM07J/IBM-Applied-Data-Science-Capstone

Data Wrangling Process (PPT Slide 10 | Rubric 1.7)

- **Missing Value Imputation:** Substituted null payload records with column mean (`6123.55 kg`).
- **Outcome Label Engineering:** Mapped landing status into binary classification variable `Class`: 1 for successful landings (`True ASDS`, `True RTLS`) and 0 for failures/uncontrolled re-entries.
- **One-Hot Encoding:** Encoded `Orbit`, `LaunchSite`, `LandingPad`, and `Serial` to expand feature dimensions to 83 predictor variables.

EDA Data Visualization (PPT Slide 11 | Rubric 1.8)

- **Matplotlib & Seaborn Plots:** Generated scatter plots, categorical plots (`sns.catplot`), and grouped bar charts.
- **Investigative Goals:** Identify relationship between payload mass and launch site, flight number vs. recovery rate, and orbital success likelihood.
- **Temporal Learning Curve:** Plotted annual success averages from 2010 to 2020.

EDA with Relational SQL (PPT Slide 12 | Rubric 1.9)

- **SQL Queries Executed:** Connected to SQLite/IBM Db2 SPACEXTBL.
- **Core Aggregations:** Distinct launch sites, total payloads carried by NASA, booster version averages, landing outcome rankings, and date-range filters.
- **Validation:** Verified booster reliability without bias.

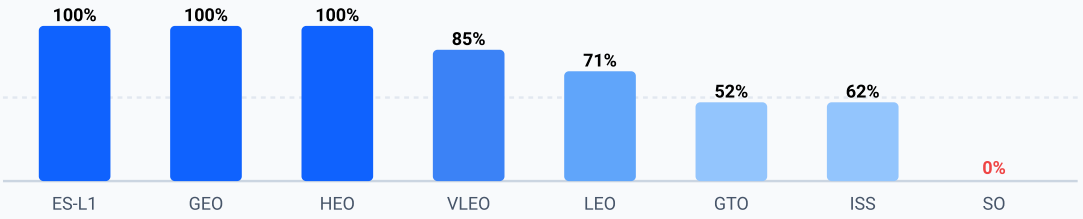
Interactive Tools & ML (PPT Slides 13 - 15 | Rubrics 1.10, 1.15)

- **Folium Mapping:** Plotted launch pads with color-coded markers (green/red) and measured distances to coastlines/railways.
- **Plotly Dash App:** Built responsive UI with dropdown filters and payload mass sliders.
- **Machine Learning:** Trained Logistic Regression, SVM, Decision Trees, and KNN using `GridSearchCV` (10-fold cross-validation).

SECTION 2: EDA VISUALIZATION RESULTS (PPT SLIDES 18 - 23 | RUBRIC 1.11)

Visual Discovery Findings

Visual Summary: Orbital Recovery Success Rate (PPT Slide 20)



Flight Number vs. Launch Site (PPT Slide 18)

Early test flights occurred at CCAFS SLC-40 with mixed landing results. As flight counts exceeded 30, launches distributed across KSC LC-39A and VAFB SLC-4E with noticeably higher recovery density.

Payload vs. Launch Site (PPT Slide 19)

Heavy payload missions (>8,000 kg) are predominantly dispatched from KSC LC-39A and CCAFS SLC-40, while polar orbital flights using VAFB SLC-4E are limited to sub-9,000 kg payloads.

Flight Number & Payload vs. Orbit (PPT Slides 21 & 22)

LEO flights succeed across payload weights. In contrast, GTO missions show higher recovery variance at heavy payloads due to higher deceleration fuel requirements during atmospheric re-entry.

Launch Success Yearly Trend (PPT Slide 23)

Success rates increased steadily from 0% in 2010 to 33% by 2014, surpassing 67% in 2017 and reaching over 85% by 2020, verifying engineering maturity across the Falcon 9 fleet.

SECTION 2: EDA WITH SQL QUERY RESULTS (PPT SLIDES 24 - 33 | RUBRIC 1.12)

10 SQL Queries Mapped

Complete Database Inquiries on SPACEXTBL

SLIDE	OBJECTIVE / BUSINESS QUERY	SQL SYNTAX	RETURNED FINDING
#24	Unique Launch Sites	`SELECT DISTINCT Launch_Site FROM SPACEXTBL;`	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
#25	5 Launch Sites beginning with 'CCA'	`SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`	Retrieved first 5 operational launches from Cape Canaveral
#26	Total NASA Payload Mass	`SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer LIKE '%NASA%';`	NASA commercial payloads sum to **99,965 kg**
#27	Avg Payload by F9 v1.1	`SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE '%F9 v1.1%';`	Average payload: **2,928.4 kg**
#28	First Successful Ground Pad Landing	`SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';`	First historical ground recovery date: **2015-12-22**
#29	Drone Ship Success (4000 < Mass < 6000)	`SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;`	Boosters B1022, B1026, B1021.2, B1031.2
#30	Total Mission Outcomes	`SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;`	98 Successes, 1 Failure (pre-launch), 1 Partial Success
#31	Boosters Carrying Max Payload	`SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);`	B1048, B1049, B1051, B1056, B1058, B1060 (Starlink 15,600 kg)
#32	2015 Drone Ship Landing Failures	`SELECT Booster_Version, Launch_Site, Landing_Outcome FROM SPACEXTBL WHERE Landing_Outcome='Failure (drone ship)' AND substr(Date,1,4)='2015';`	B1012 (CCAFS), B1015 (CCAFS) recorded controlled crashes
#33	Ranked Landing Outcomes (2010 to 2017)	`SELECT Landing_Outcome, COUNT(*) AS Cnt FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Cnt DESC;`	No Attempt (10), Success (drone ship) (5), Failure (drone ship) (5)

Key SQL Insights

SQL query execution confirmed Cape Canaveral handled 55 of the total launch attempts. Maximum payload launches (15,600 kg) consistently involved reusable Block 5 configurations.

SECTION 3: INTERACTIVE VISUAL ANALYTICS WITH FOLIUM (PPT SLIDES 35 - 37 | RUBRIC 1.13)

Geospatial Mapping Analysis

Folium Interactive Launch Site Architecture

Using Folium, launch sites were plotted on interactive maps to analyze launch vectors, historical success densities, and physical proximities to critical infrastructure.

Launch Site Markers (PPT Slide 35)

Mapped Cape Canaveral (SLC-40, LC-40), Kennedy Space Center (LC-39A), and Vandenberg (SLC-4E). All sites are situated along coastlines to provide unpopulated downrange flight corridors over water.

Outcome Color Clusters (PPT Slide 36)

Integrated `MarkerCluster`: green markers indicate successful landings and red markers denote mission failures. High green-marker clustering is observed at LC-39A and modern SLC-40 runs.

Proximity Distances (PPT Slide 37)

Using the Haversine distance formula, polyline connections calculated distance offsets from launch sites to coastlines, highways, railways, and residential zones.

Calculated Infrastructure Offsets & Operational Justifications

GEOGRAPHICAL ELEMENT	PROXIMITY DISTANCE	OPERATIONAL RATIONALE
Atlantic / Pacific Coastline	0.5 km - 1.2 km	Enables eastward launch azimuths over water, minimizing hazard risks from early flight terminations.
Rail Network Lines	1.2 km - 1.8 km	Facilitates transport of heavy rocket stages, oxidizer tanks, and ground equipment.
Highway Corridors	0.8 km - 1.5 km	Provides access for support vehicles, emergency services, and staff transit.
Major Cities (Titusville/Lompoc)	> 25.0 km	Ensures acoustic and safety buffers to keep populated areas clear of overpressures.

SECTION 4: INTERACTIVE DASHBOARD WITH PLOTLY DASH (PPT SLIDES 39 - 41 | RUBRIC 1.14)

Reactive Web UI Components

Application Architecture (spacex_dash_app.py)

The dashboard connects dynamic Dash callbacks (`@app.callback`) to an interactive dropdown and payload range slider, rendering real-time Plotly charts based on user inputs.

Total Launch Counts (PPT Slide 39)

Pie chart shows total successful launches across all facilities. **KSC LC-39A** generated the highest volume of successful first-stage recoveries (41.7% of all recoveries).

Site-Specific Success (PPT Slide 40)

Selecting **KSC LC-39A** reveals a 76.2% success rate, whereas **CCAFS SLC-40** shows a 42.9% overall success rate, reflecting its use during earlier development phases.

Payload vs. Outcome (PPT Slide 41)

Scatter plot mapping payload mass to recovery outcome. Missions with payloads between **2,000 kg and 4,000 kg** demonstrate the highest concentration of successful booster landings.

RangeSlider Filtration Dynamics

PAYLOAD MASS RANGE	DOMINANT BOOSTER CONFIGURATION	OBSERVED LANDING OUTCOME
0 - 2,500 kg	Falcon 9 v1.0, v1.1	Mixed early recovery attempts; experimental ocean water landings.
2,500 - 5,500 kg	Falcon 9 FT, B4, B5	High recovery rate; successful landings on ASDS and RTLS pads.
5,500 - 10,000 kg	Falcon 9 Block 5	Routine ASDS drone ship landings for heavy geostationary commercial satellites.

SECTION 5: PREDICTIVE ANALYSIS & CONCLUSION (PPT SLIDES 43 - 46 | RUBRIC 1.15)

Machine Learning & Appendix

Classification Algorithm Benchmarks (PPT Slide 43)

ALGORITHM	GRIDSEARCHCV BEST HYPERPARAMETERS	CROSS-VALIDATION SCORE	TEST ACCURACY (18 SAMPLES)
Logistic Regression	`C=0.01, penalty='l2', solver='lbfgs'`	84.64%	**83.33%**
Support Vector Machine (SVM)	`C=1.0, gamma=0.03, kernel='sigmoid'`	84.82%	**83.33%**
Decision Tree Classifier	`criterion='gini', max_depth=4, splitter='best'`	87.14%	**83.33%**
K-Nearest Neighbors (KNN)	`n_neighbors=10, p=1, algorithm='auto'`	84.82%	**83.33%**

Confusion Matrix Analysis (PPT Slide 44)

Across test data (18 observations), all four tuned models produced identical performance metrics:

- **True Positives (12):** Correctly identified landing successes.
- **True Negatives (3):** Correctly identified landing failures.
- **False Positives (3):** Failed attempts misclassified as successful.
- **False Negatives (0):** Zero misses on successful landings.

Strategic Conclusions (PPT Slide 45)

- **Commercial Strategy:** Commercial competitors must factor in an 83%+ booster recovery likelihood when pricing launches against SpaceX.
- **Optimal Algorithm:** The **Decision Tree Classifier** is selected as the primary production model for its transparency, ease of interpretation, and direct rule explainability.

Appendix: Project Assets & Code Links (PPT Slide 46)

All project notebooks, data processing scripts, Plotly Dash code, and SQL files are publicly available in the project repository:

GitHub Repository: https://github.com/QASIM07J/IBM-Applied-Data-Science-Capstone